

Question 1: PROGRAM TO CALCULATE SIMPLE INTEREST.

```
#include<stdio.h>

int main(){

int principle,rate,time;

printf("Enter the principle amount:");

scanf("%d",&principle);

printf("Enter the rate:");

scanf("%d",&rate);

printf("Enter the period:");

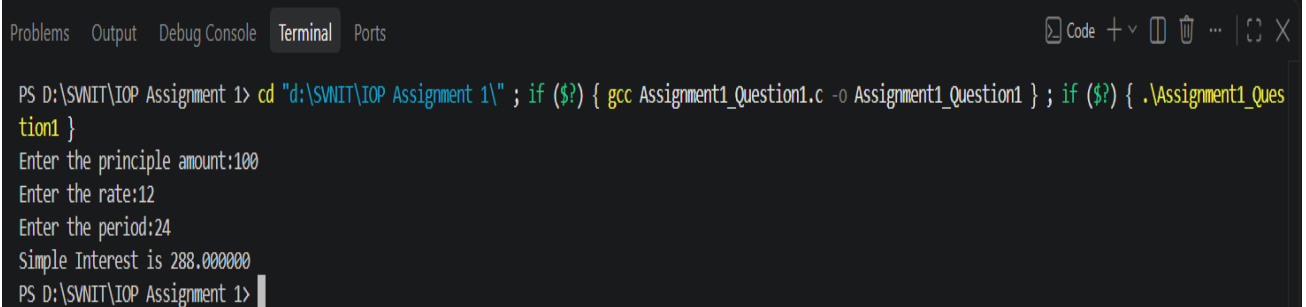
scanf("%d",&time);

double SI;

SI=(principle*rate*time)/100;

printf("Simple Interest is %Lf",SI);

}
```



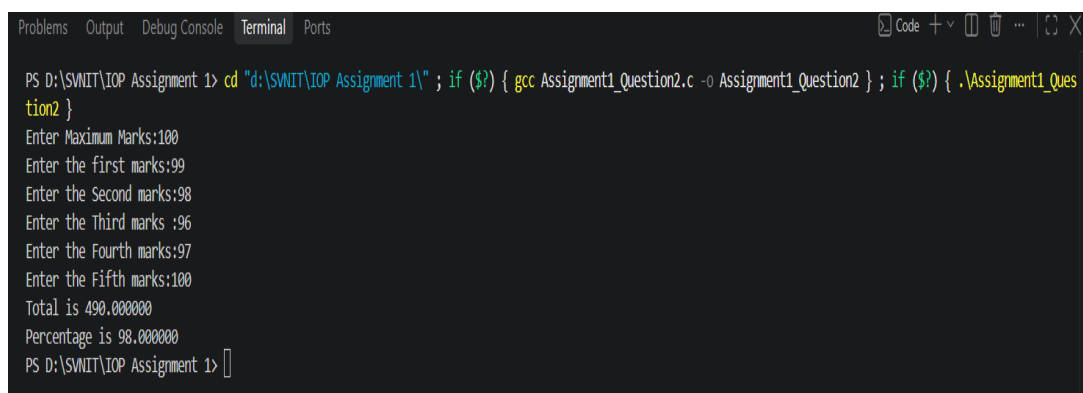
The screenshot shows a Windows Command Prompt window with the following content:

```
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question1.c -o Assignment1_Question1 } ; if ($?) { .\Assignment1_Question1 }
Enter the principle amount:100
Enter the rate:12
Enter the period:24
Simple Interest is 288.000000
PS D:\SVNIT\IOP Assignment 1>
```

The terminal output demonstrates the program's execution. It prompts the user for the principle amount (100), rate (12), and period (24). The calculated simple interest is displayed as 288.000000.

Question 2: PROGRAM TO READ MARKS OF FIVE SUBJECT OF A STUDENT AND CALCULATE TOTAL AND PERCENTAGE.

```
#include<stdio.h>
int main(){
    int m1,m2,m3,m4,m5,max,base;
    double per,total;
    printf("Enter Maximum Marks:");
    scanf("%d",&max);
    printf("Enter the first marks:");
    scanf("%d",&m1);
    printf("Enter the Second marks:");
    scanf("%d",&m2);
    printf("Enter the Third marks :");
    scanf("%d",&m3);
    printf("Enter the Fourth marks:");
    scanf("%d",&m4);
    printf("Enter the Fifth marks:");
    scanf("%d",&m5);
    base=max;
    total=m1+m2+m3+m4+m5;
    per=(total/(base*5))*100;
    printf("Total is %Lf\n",total);
    printf("Percentage is %Lf\n",per);
}
```



```
Problems Output Debug Console Terminal Ports
PS D:\SWUIT\IOP Assignment 1> cd "d:\SWUIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question2.c -o Assignment1_Question2 } ; if ($?) { .\Assignment1_Question2 }
Enter Maximum Marks:100
Enter the first marks:99
Enter the Second marks:98
Enter the Third marks :96
Enter the Fourth marks:97
Enter the Fifth marks:100
Total is 490.000000
Percentage is 98.000000
PS D:\SWUIT\IOP Assignment 1>
```

Question-3: PROGRAM TO CALCULATE GROSS SALARY

```
#include<stdio.h>

int main(){

    int basicsal;

    double house_rent_allowance,daily_allowance,PF,grosssal;

    printf("Enter your Basic Salary : ");

    scanf("%d",&basicsal);

    house_rent_allowance=basicsal*0.20;

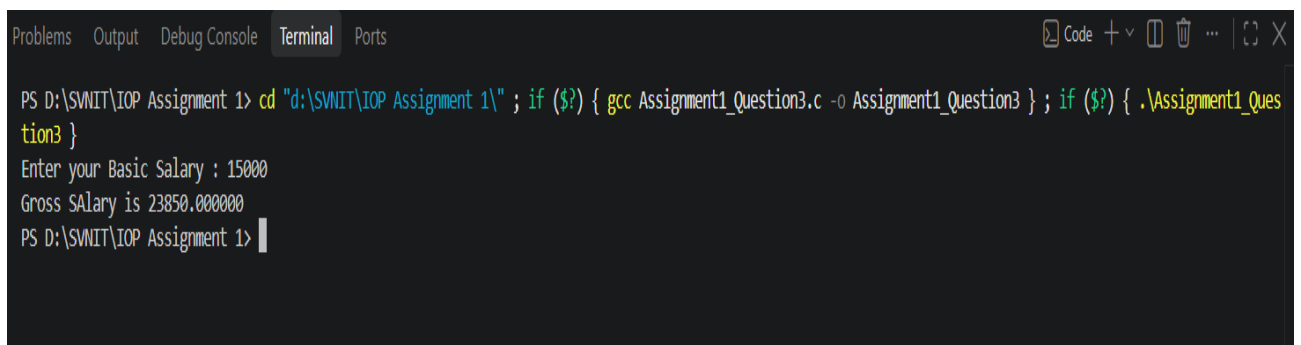
    daily_allowance=basicsal*0.50;

    PF=basicsal*0.11;

    grosssal=basicsal+house_rent_allowance+daily_allowance-PF;

    printf("Gross SALary is %Lf",grosssal);

}
```



The screenshot shows a Windows Command Prompt window with the following text:

```
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question3.c -o Assignment1_Question3 } ; if ($?) { .\Assignment1_Question3 }
Enter your Basic Salary : 15000
Gross SALary is 23850.000000
PS D:\SVNIT\IOP Assignment 1>
```

The terminal output shows the program successfully compiled and executed, taking the input '15000' and producing the output 'Gross SALary is 23850.000000'.

Question-4: PROGRAM TO CONVERT TEMPERATURE FROM FAHRENHEIT TO CENTIGRADE DEGREES.

```
#include<stdio.h>

int main(){

    float temp;

    double cel;

    printf("Enter temperature in Fahrenheit:");

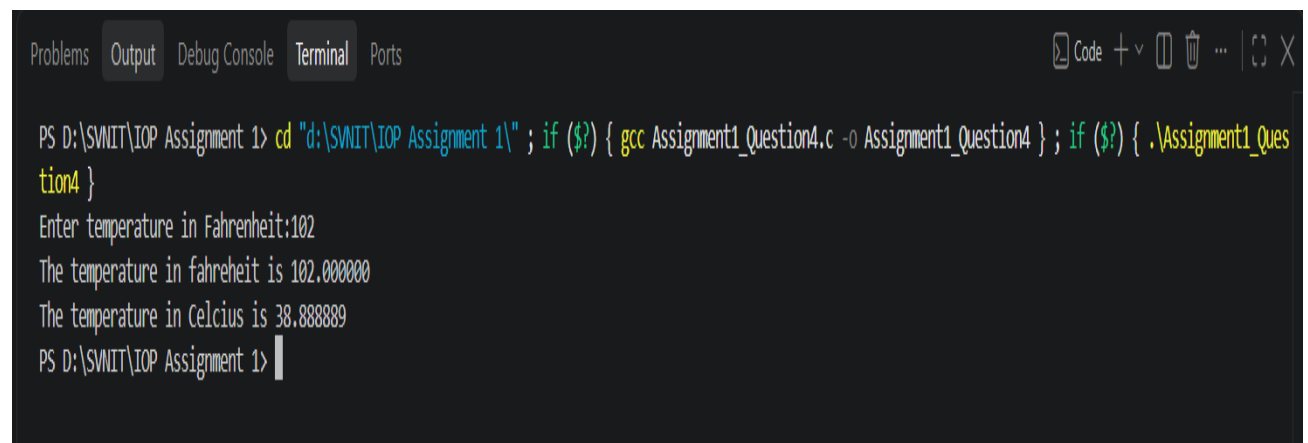
    scanf("%f",&temp);

    cel=(temp-32)/1.8;

    printf("The temperature in fahreheit is %f\n",temp);

    printf("The temperature in Celcius is %Lf\n",cel);

}
```



The screenshot shows a terminal window with a dark background. At the top, there are tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. The terminal displays the following text:

```
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question4.c -o Assignment1_Question4 } ; if ($?) { .\Assignment1_Question4 }
Enter temperature in Fahrenheit:102
The temperature in fahreheit is 102.000000
The temperature in Celcius is 38.888889
PS D:\SVNIT\IOP Assignment 1> |
```

Question -5: PROGRAM TO SWAP TWO VARIABLES USING THIRD VARIABLE.

```
#include<stdio.h>

int main(){

    int a,b,temp;

    printf("Enter the first Variable:");

    scanf("%d",&a);

    printf("Enter the second Variable:");

    scanf("%d",&b);

    temp=a;

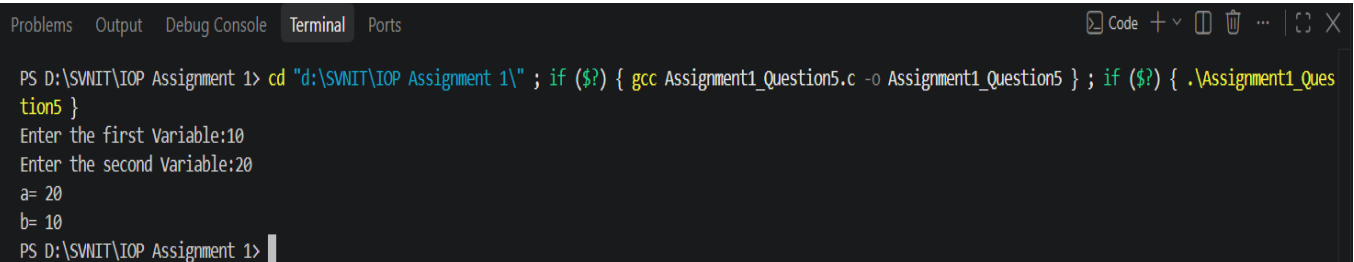
    a=b;

    b=temp;

    printf("a= %d\n",a);

    printf("b= %d\n",b);

}
```



The screenshot shows a Windows Command Prompt window with the following text:

```
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question5.c -o Assignment1_Question5 } ; if ($?) { .\Assignment1_Question5 }
Enter the first Variable:10
Enter the second Variable:20
a= 20
b= 10
PS D:\SVNIT\IOP Assignment 1>
```

The terminal output demonstrates the program's execution. It prompts the user to enter two variables. The first variable is 10, and the second variable is 20. The program then prints the values of the variables after swapping, showing 'a= 20' and 'b= 10'.

Question 6: PROGRAM TO SWAP TWO VARIABLES WITHOUT USING THIRD VARIABLE.

```
#include<stdio.h>

int main(){

    int a,b;

    printf("Enter the first Variable:");

    scanf("%d",&a);

    printf("Enter the second Variable:");

    scanf("%d",&b);

    a=a-b;

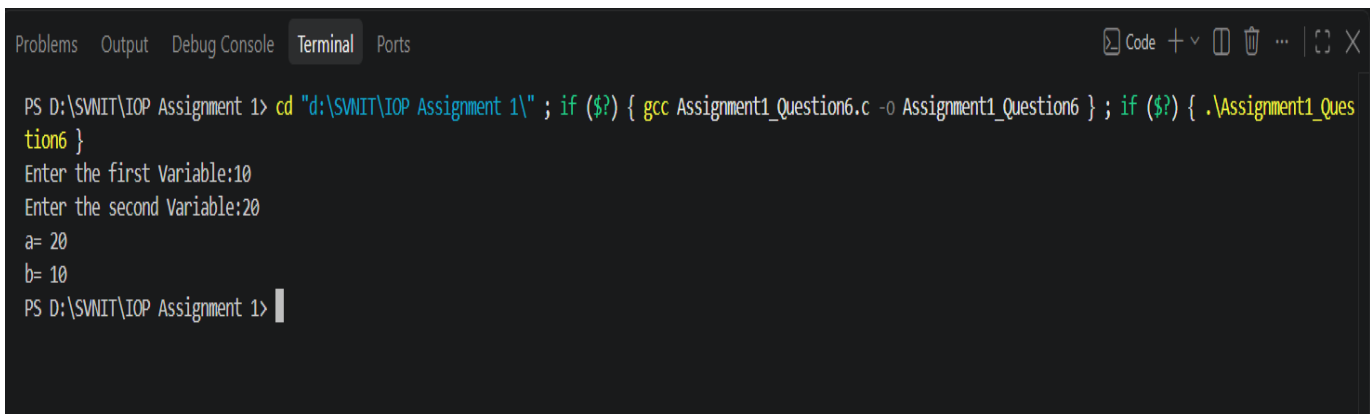
    b=a+b;

    a=b-a;

    printf("a= %d\n",a);

    printf("b= %d\n",b);

}
```



The screenshot shows a terminal window with the following content:

```
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question6.c -o Assignment1_Question6 } ; if ($?) { .\Assignment1_Question6 }
Enter the first Variable:10
Enter the second Variable:20
a= 20
b= 10
PS D:\SVNIT\IOP Assignment 1>
```

The terminal output demonstrates the program's execution. It prompts for the first variable (10) and the second variable (20). After running the program, it displays the swapped values: a= 20 and b= 10.

Question 7: PROGRAM TO CALCULATE AREA OF A TRIANGLE.

```
#include<stdio.h>

int main(){

    int b,h;

    float area;


    printf("Enter the height of the triangle:");

    scanf("%d",&h);

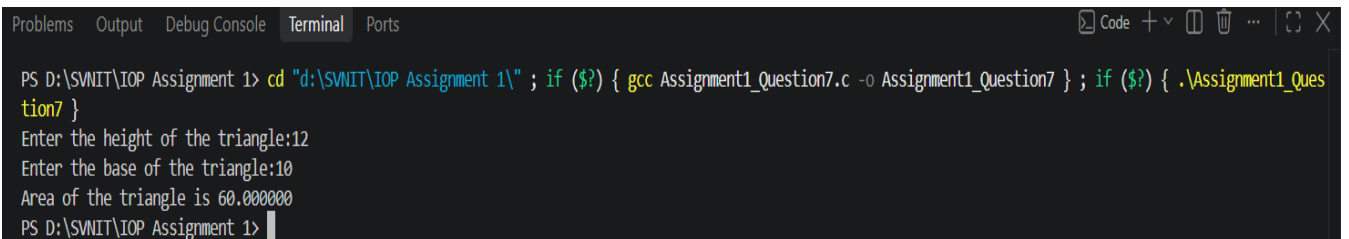
    printf("Enter the base of the triangle:");

    scanf("%d",&b);

    area=0.5*b*h;

    printf("Area of the triangle is %f",area);

}
```



The screenshot shows a terminal window with the following content:

```
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question7.c -o Assignment1_Question7 } ; if ($?) { .\Assignment1_Question7 }
Enter the height of the triangle:12
Enter the base of the triangle:10
Area of the triangle is 60.000000
PS D:\SVNIT\IOP Assignment 1> |
```

The terminal window has tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. The command prompt shows the directory path 'PS D:\SVNIT\IOP Assignment 1' and the execution of a C program that calculates the area of a triangle with height 12 and base 10, resulting in an area of 60.000000.

Question 8: EARTH TAKES A PERIOD OF REVOLUTION OF 31558150 SECONDS .WRITE A C PROGRAM TO CONVERT THIS INTO NUMBER OF DAY,HOURS AND MINUTES

```
#include<stdio.h>

int main(){

    double hour,day,min;

    int rem1,rem2;


    printf("Earth takes a revolution of 31558150 seconds so converting it into day,
hour and min\n");

    day=31558150/86400;

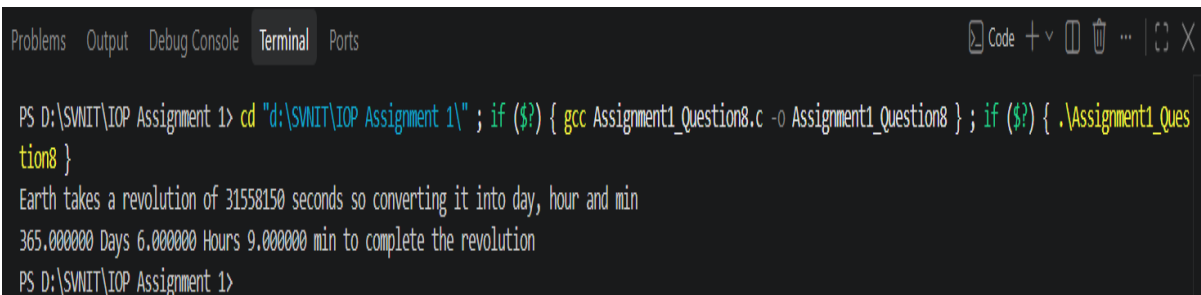
    rem1=31558150%86400;

    hour=rem1/3600;

    rem2=rem1%3600;

    min=rem2/60;

    printf("%Lf Days %Lf Hours %Lf min to complete the revolution",day,hour,min);
}
```



```
Problems Output Debug Console Terminal Ports
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question8.c -o Assignment1_Question8 } ; if ($?) { .\Assignment1_Question8 }
Earth takes a revolution of 31558150 seconds so converting it into day, hour and min
365.000000 Days 6.000000 Hours 9.000000 min to complete the revolution
PS D:\SVNIT\IOP Assignment 1>
```

Question 9: PROGRAM TO READ TIME IN HR,MIN,SEC AND CONVERT IT INTO TOTAL SECOND.

```
#include<stdio.h>

int main(){

    int hour, min, sec;

    int hourtosec,mintosec,totalsec;


    printf("Enter time in hour:min:sec format");

    scanf("%d : %d : %d",&hour,&min,&sec);

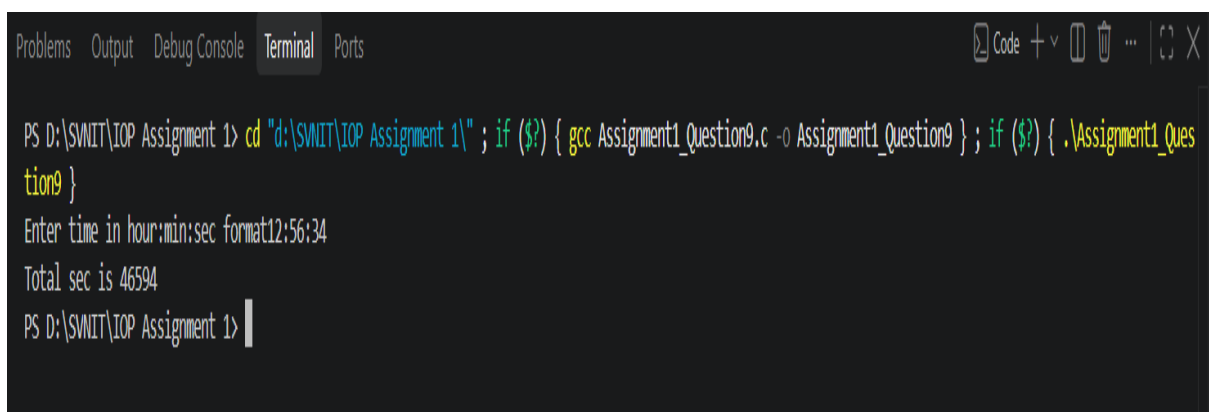

    hourtosec=hour*60*60;

    mintosec=min*60;

    totalsec=hourtosec+mintosec+sec;

    printf("Total sec is %d",totalsec);

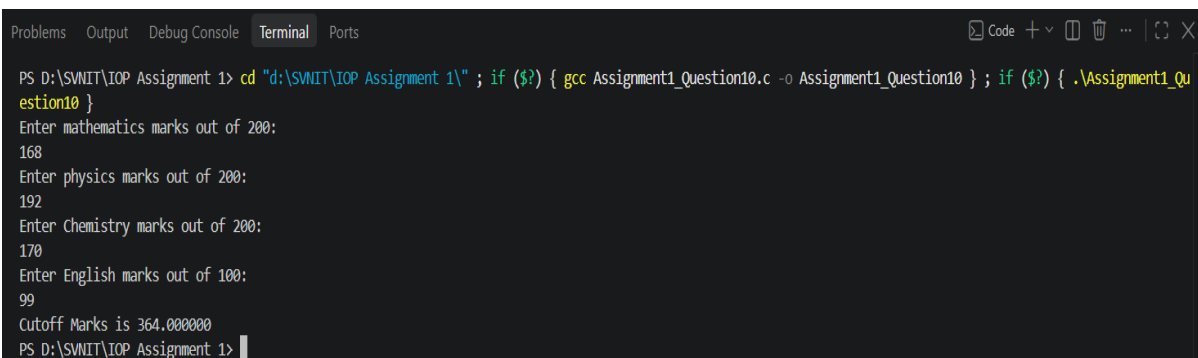
}
```



```
Problems Output Debug Console Terminal Ports
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question9.c -o Assignment1_Question9 } ; if ($?) { .\Assignment1_Question9 }
Enter time in hour:min:sec format12:56:34
Total sec is 46594
PS D:\SVNIT\IOP Assignment 1> |
```

Question 10: WRITE A C PROGRAM TO CALCULATE THE CUTOFF MARK OF A STUDENT USING THE FORMULA. $CM = M/2 + P/2 + C/2 + E$ WHERE CM = Cut of f mark M = Marks in Mathematics out of 200 P = Marks in Physics out of 200 C = Marks in Chemistry out of 200 E = Marks in entrance examination out of 100

```
#include<stdio.h>
int main(){
    int M,P,C,E;
    float CM;
    printf("Enter mathematics marks out of 200:\n");
    scanf("%d",&M);
    printf("Enter physics marks out of 200:\n");
    scanf("%d",&P);
    printf("Enter Chemistry marks out of 200:\n");
    scanf("%d",&C);
    printf("Enter English marks out of 100:\n");
    scanf("%d",&E);
    CM=(M/2)+(P/2)+(C/2)+E;
    printf("Cutoff Marks is %f",CM);
}
```



The screenshot shows a terminal window with the following content:

```
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question10.c -o Assignment1_Question10 } ; if ($?) { .\Assignment1_Question10 }
Enter mathematics marks out of 200:
168
Enter physics marks out of 200:
192
Enter Chemistry marks out of 200:
170
Enter English marks out of 100:
99
Cutoff Marks is 364.000000
PS D:\SVNIT\IOP Assignment 1>
```

Question 11: PROGRAM TO READ TOTAL SECOND AND CONVERT IT INTO TIME

```
#include<stdio.h>

int main(){

    double hour,sec,min;

    int totalsec;


    printf("Enter Total Second\n");

    scanf("%d",&totalsec);

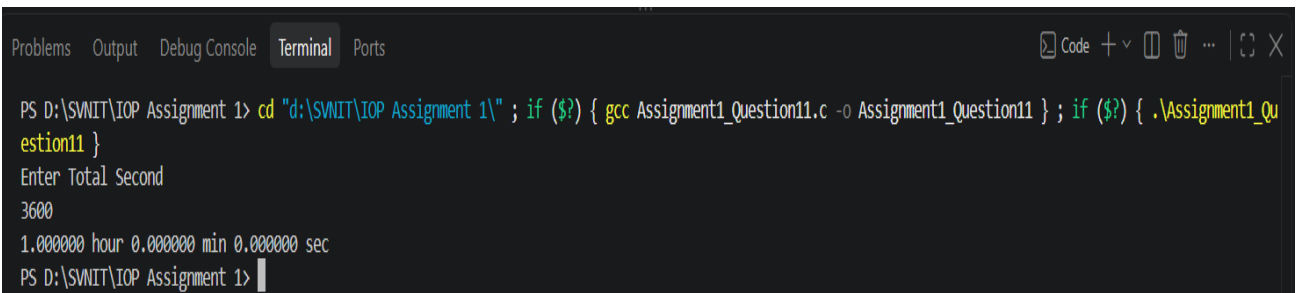
    hour=totalsec/3600;

    min=(totalsec%3600)/60;

    sec=totalsec%60;

    printf("%Lf hour %Lf min %Lf sec",hour,min,sec);

}
```



```
Problems Output Debug Console Terminal Ports
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question11.c -o Assignment1_Question11 } ; if ($?) { .\Assignment1_Question11 }
Enter Total Second
3600
1.000000 hour 0.000000 min 0.000000 sec
PS D:\SVNIT\IOP Assignment 1> |
```

